

NIVELL 1: Creació BBDD

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.

Creació de la BBDD *Sprint4* :

```
SQL File 7*
1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1*/
2 /* Dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.*/
3 /* Creo la BBDD Sprint4.*/
4 CREATE DATABASE IF NOT EXISTS Sprint4;
5 USE Sprint4;
```

Creació de la taula de dimensions *users*:

```
ver Tools Scripting Help
SQL_creacio_BBDD_mes_taula... SQL_creacio_taula_european_u... SQL_creacio_taula_companies SQL File 11* SQL File 12*
1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1*/
2 /* Dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.*/
3 /* Creo la taula de dimensions users on afegire els registres american_user i els european_user.*/
4 CREATE TABLE IF NOT EXISTS users (
5     id CHAR(10) PRIMARY KEY,
6     name VARCHAR(100),
7     surname VARCHAR(100),
8     phone VARCHAR(150),
9     email VARCHAR(150),
10    birth_date VARCHAR(100),
11    country VARCHAR(150),
12    city VARCHAR(150),
13    postal_code VARCHAR(100),
14    address VARCHAR(255)
15 );

Output
Action Output
# Time Action Message Duration / Fetch
1 12:21:11 CREATE TABLE IF NOT EXISTS users (id CHAR(10) PRIMARY KEY, name VARCHAR(100), surname VAR... 0 row(s) affected 0.047 sec
```

Omplir la taula *users* usant els fitxers *american_users.csv* i *european_users.csv*:

The screenshot shows the SQL Server Enterprise Manager interface. The SQL File 26 window contains the following commands:

```
1 LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\european_users.csv'
2 INTO TABLE users
3 FIELDS TERMINATED BY ','
4 ENCLOSED BY '"'
5 IGNORE 1 LINES;
6
7 LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv'
8 INTO TABLE users
9 FIELDS TERMINATED BY ','
10 ENCLOSED BY '"'
11 IGNORE 1 LINES;
12
13 select * from users;
```

The Result Grid shows the data loaded from the CSV files:

id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagitts St.
2	Garrett	Mcconnell	(718) 257-2412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave
3	Ciaran	Harrison	(522) 598-1365	interdum.feugiat@aol.org	Apr 29, 1998	United States	Houston	77001	736-2063 Tellus St.
4	Howard	Stafford	1-411-740-3269	ornare.egestas@icloud.edu	Feb 18, 1989	United States	Phoenix	85001	Ap #545-2244 Erat. Rd.
5	Hayfa	Pierce	1-554-541-2077	et.malesuada.fames@hotmail.org	Sep 26, 1998	United States	Philadelphia	19101	341-2821 Ultrices Av.

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:47:56	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\european_users.csv' INTO...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0	0.157 sec
2	14:48:47	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\american_users.csv' INTO...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.031 sec
3	14:48:57	select * from users	5000 row(s) returned	0.000 sec / 0.016 sec

Creació de al taula de dimensions *companies*:

The screenshot shows the SQL File 10 window containing the following command:

```
1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1 */
2 /* Dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.*/
3 /* Creo la taula de dimensions companies.*/
4 CREATE TABLE IF NOT EXISTS companies (
5     company_id CHAR(10) PRIMARY KEY,
6     company_name VARCHAR(250),
7     phone VARCHAR(150),
8     email VARCHAR(150),
9     country VARCHAR(150),
10     website VARCHAR(150)
11 );
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	12:08:26	CREATE TABLE IF NOT EXISTS companies (company_id CHAR(10) PRIMARY KEY, company_name VAR...	0 row(s) affected	0.031 sec

Omplir la taula *companies* usant els fitxers *companies.csv*:

The screenshot shows the SQL Server Enterprise Manager interface. The SQL script in the editor is as follows:

```
1 LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv'  
2 INTO TABLE companies  
3 FIELDS TERMINATED BY ','  
4 IGNORE 1 LINES;  
5  
6 select * from companies;
```

The 'Result Grid' displays the data loaded from the CSV file:

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
b-2234	Conville In Incorporated	06 66 57 28 50	mauris.ut@aol.co.uk	Germany	https://cm.com/user/110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.prim@outlook.ca	New Zealand	https://netflix.com/settings
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.co.uk	Norway	https://nytimes.com/user/110

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	17:45:08	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\companies.csv' INTO TA...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.000 sec
2	17:45:15	select * from companies	100 row(s) returned	0.000 sec / 0.000 sec

Creació taula de dimensions *products*:

The screenshot shows the SQL Server Enterprise Manager interface. The SQL script in the editor is as follows:

```
1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1 */  
2 /* Dissenyar una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.*/  
3 /* Crear la taula de products.*/  
4 CREATE TABLE IF NOT EXISTS products (  
5     id int PRIMARY KEY,  
6     product_name VARCHAR(150),  
7     price decimal(10,2),  
8     colour VARCHAR(10),  
9     weight decimal(5,1),  
10    birth_date VARCHAR(100),  
11    warehouse_id VARCHAR(50)  
12 );
```

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	12:41:28	CREATE TABLE IF NOT EXISTS products (id int PRIMARY KEY, product_name VARCHAR(150), price deci...	0 row(s) affected	0.047 sec

Omplir la taula *products* usant el fitxer *products.csv*:

The screenshot shows the SQL Server Enterprise Manager interface. The script being executed is as follows:

```

1 LOAD DATA INFILE 'C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\products.csv'
2 INTO TABLE products
3 FIELDS TERMINATED BY ','
4 IGNORE 1 LINES
5 (@id, @product_name, @raw_price, @colour, @weight, @warehouse_id)
6 SET
7 id = @id,
8 product_name = TRIM(@product_name),
9 price = CAST(SUBSTRING(@raw_price, 2) AS DECIMAL(10,2)),
10 colour = @colour,
11 weight = @weight,
12 warehouse_id = @warehouse_id;
13
14 select * from products;

```

The Result Grid shows the following data:

id	product_name	price	colour	weight	warehouse_id
1	Direwolf Stannis	161.11	#7c7c7c	1.0	WH-4
2	Tarly Stark	9.24	#919191	2.0	WH-3
3	duel tourney Lannister	171.13	#d8d8d8	1.5	WH-2
4	warden south duel	71.89	#111111	3.0	WH-1
5	skywalker ewok	171.22	#dbdbdb	3.2	WH-0

The Output pane shows the execution log:

```

# Time Action Message Duration / Fetch
1 18:08:25 LOAD DATA INFILE 'C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\products.csv' INTO TABLE... 100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0 0.016 sec
2 18:08:59 select * from products 100 row(s) returned 0.000 sec / 0.000 sec

```

Dono format a la columna *price* per tal d'eliminar el caràcter inicial \$ i convertir-la a decimal en el moment d'introduir el valor a la taula.

Creació taula de dimensions *credit_card*:

The screenshot shows the SQL Server Enterprise Manager interface. The script being executed is as follows:

```

1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1*/
2 /* Dissenyar una base de dades amb un esquema d'estrella que contingui, almenys 4 taules.*/
3 /* Crea la taula de dimensions credit_card amb FK que apunta a users.*/
4 CREATE TABLE IF NOT EXISTS credit_card (
5 id CHAR(20) PRIMARY KEY,
6 user_id CHAR(10),
7 iban VARCHAR(50),
8 pan VARCHAR(50),
9 pin VARCHAR(4),
10 cvv int,
11 track1 VARCHAR(150),
12 track2 VARCHAR(150),
13 expiring_date date,
14 FOREIGN KEY (user_id) REFERENCES users(id)
15 );

```

The Output pane shows the execution log:

```

# Time Action Message Duration / Fetch
1 12:29:14 CREATE TABLE IF NOT EXISTS credit_card (id CHAR(20) PRIMARY KEY, user_id CHAR(10), iban VAR... 0 row(s) affected 0.047 sec

```

Omplir la taula *credit_card* usant el fitxer *credits_cards.csv*:

The screenshot shows the SQL Server Enterprise Manager interface. The script being executed is as follows:

```

1 LOAD DATA INFILE 'C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\credits_cards.csv'
2 INTO TABLE credit_card
3 FIELDS TERMINATED BY ','
4 IGNORE 1 LINES
5 (@id, @user_id, @iban, @pan, @pin, @cvv, @track1, @track2, @expiring_date)
6 SET
7   id = @id,
8   user_id = @user_id,
9   iban = @iban,
10  pan = @pan,
11  pin = @pin,
12  cvv = @cvv,
13  track1 = @track1,
14  track2 = @track2,
15  expiring_date = STR_TO_DATE(@expiring_date, '%m/%d/%y');
16
17 select * from credit_card;

```

The Result Grid shows the following data:

	id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date
CcS-4857	276	XX4857591835292505850771	2314242385113924	1819	467	%82314242385113924~LWCBLUWCBUD~22...	%82314242385113924~2410101518363164~	2025-09-27	
CcS-4858	277	XX8581768137002436094025	6582720299715533	3964	817	%86582720299715533~TIQMVITIQMVI~2404...	%86582720299715533~2411101104546272~	2028-12-28	
CcS-4859	278	XX7826930491423553609370	8861684536289642	4983	277	%88861684536289642~COFBDGCOFBDG~280...	%88861684536289642~2502101761665371~	2026-11-26	
CcS-4860	279	XX555990368835304645299	2481155515498459	6876	661	%82481155515498459~TIJUTUTIJUTU~31040...	%82481155515498459~2602101514414395~	2027-07-27	
CcS-4861	280	XX203518287719191627307	1308930301149557	5710	398	%81308930301149557~HPOBNZHPBNZ~330...	%81308930301149557~2805101751305028~	2026-04-25	

The Output pane shows the following messages:

```

1 18:49:50 LOAD DATA INFILE 'C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\credits_cards.csv' INTO T... 5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0 0.422 sec
2 18:50:01 select * from credit_card 5000 row(s) returned 0.000 sec / 0.032 sec

```

Dono format a la columna *expiring_date* per tal convertir-la a valor tipus *date* en el moment d'introduir el valor a la taula.

Creació taula de fets (hechos) *transactions*:

The screenshot shows the SQL Server Enterprise Manager interface. The script being executed is as follows:

```

1 /* Sprint 4: consulta nivell 1 - exercici 1 - punt 1 */
2 /* Dissenyar una base de dades amb un esquema d'estrelles que contingui, almenys 4 taules. */
3 /* Crear la taula de dimensions transactions amb les seves FK que apunten a varies taules. */
4 CREATE TABLE IF NOT EXISTS transactions (
5   id CHAR(50) PRIMARY KEY,
6   card_id CHAR(20),
7   business_id CHAR(10),
8   timestamp timestamp,
9   amount DECIMAL(10,2),
10  declined boolean,
11  product_id CHAR(50),
12  user_id CHAR(10),
13  lat float,
14  longitude float,
15  FOREIGN KEY (card_id) REFERENCES credit_card(id),
16  FOREIGN KEY (business_id) REFERENCES companies(company_id),
17  FOREIGN KEY (user_id) REFERENCES users(id)
18 );

```

The Output pane shows the following message:

```

1 18:10:46 CREATE TABLE IF NOT EXISTS transactions (id CHAR(50) PRIMARY KEY, card_id CHAR(20), busine... 0 row(s) affected 0.078 sec

```

Omplir la taula *transactions* usant el fitxer *transactions.csv*:

The screenshot shows the SQL Server Enterprise Manager interface. The SQL script in the editor is as follows:

```
1 LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv'
2 INTO TABLE transactions
3 FIELDS TERMINATED BY ','
4 IGNORE 1 LINES;
5
6 select * from transactions;
```

The 'Result Grid' displays the first few rows of the 'transactions' table:

id	card_id	business_id	timestamp	amount	declined	product_ids	user_id	lat	longitude
00043A49-2949-b94B-A5DD-A5BAE38B190D	Ccs-9294	b-2458	2024-08-28 07:16:46	395.43	0	16, 26, 97, 87	4713	46.1999	1.43554
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	Ccs-9019	b-2370	2016-12-21 20:07:18	155.63	0	66, 69, 87	438	41.9972	12.2218
00045D68-ED2E-4F2F-8186-CE074D875D0	Ccs-6699	b-2390	2020-07-14 15:37:45	326.01	0	30, 11, 16, 81	2118	29.7573	-95.3796
000481C3-1C26-4FEF-83A0-4C0DEB00468D	Ccs-6696	b-2230	2017-09-04 19:44:53	161.60	0	72	2115	53.5489	-113.503
00051AA4-9CDE-4268-B070-C38062A1B3E2	Ccs-7606	b-2266	2017-01-05 18:19:25	148.91	0	18	3025	52.2084	5.69081

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	18:58:05	LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\transactions.csv' INTO TA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0	10.000 sec
2	18:58:23	select * from transactions	100000 row(s) returned	0.015 sec / 0.219 sec

NIVELL 1: Exercici 1

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

The screenshot shows the SQL script in the editor:

```
1 /* Sprint 4: consulta nivell 1 - exercici 1 */
2 /* crear subconsulta que mostri tots els usuaris amb més de 80 transaccions */
3
4 select u.name, u.surname, count(t.id) as nro_trans from users u
5 inner join transactions t on t.user_id = u.id
6 group by name, surname
7 having nro_trans > 80
```

The 'Result Grid' displays the results of the query:

name	surname	nro_trans
Molly	Gilliam	110
Dxxwgi	Hwcdru	94
Bnyr	Astuw	91
Sfzzoh	Xgvfridx	81

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	10:30:48	select u.name, u.surname, count(t.id) as nro_trans from users u inner join transactions t on ...	4 row(s) returned	0.156 sec / 0.000 sec

NIVELL 1: Exercici 2

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia *Donec Ltd*, utilitza almenys 2 taules.

The screenshot shows a SQL IDE window titled 'consulta_nivell1_exercici1' with a 'SQL File 10*' tab. The query is as follows:

```
1 /* Sprint 4: consulta nivell 1 - exercici 2 */
2 /* Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia "Donec Ltd" */
3 • select cc.iban, round(avg(t.amount),2) as mitjana from transactions t
4 inner join credit_card cc on cc.id = t.card_id
5 inner join companies c on c.company_id = t.business_id
6 group by cc.iban, c.company_name
7 having c.company_name = 'Donec Ltd'
8 order by mitjana desc
```

The 'Result Grid' shows the following data:

iban	mitjana
XX383017813919620199366352	680.69
XX637706357397570394973913	680.01
XX971393971465292202312259	645.46
XX171847116028802375969307	628.89
XX225424638818542406223575	608.68
XX748890729057195711766071	607.29

The 'Output' section shows the execution details:

#	Time	Action	Message	Duration / Fetch
1	10:45:40	select cc.iban, round(avg(t.amount),2) as mitjana from transactions t inner join credit_card ...	371 row(s) returned	0.796 sec / 0.000 sec

NIVELL 2: Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les últimes tres transaccions van ser declinades:

```

1 CREATE TABLE credit_card_status AS (
2   WITH ultimas_transacciones AS (
3     SELECT id, card_id, declined,
4           ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS rn
5     FROM transactions
6   ),
7   top3_transacciones AS (
8     SELECT * FROM ultimas_transacciones
9     WHERE rn <= 3
10  ),
11  estado_tarjetas AS (
12    SELECT card_id, COUNT(*) AS total, SUM(declined) AS total_declined
13    FROM top3_transacciones GROUP BY card_id
14  )
15  SELECT cc.id as card_id,
16         CASE
17           WHEN et.total = 3 AND et.total_declined = 3 THEN 1 /*desactivada*/
18           ELSE 0 /*activada*/
19         END AS status
20  FROM credit_card cc
21  LEFT JOIN estado_tarjetas et ON cc.id = et.card_id
22 );
23 /*afegim clau foranea */
24 ALTER TABLE credit_card_status ADD CONSTRAINT fk_credit_card FOREIGN KEY (card_id) REFERENCES credit_card(id)
25 ON DELETE CASCADE ON UPDATE CASCADE;

```

#	Time	Action	Message	Duration / Fetch
1	12:28:21	CREATE TABLE credit_card_status AS (WITH ultimas_transacciones AS (SELECT ...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.703 sec
2	12:28:22	ALTER TABLE credit_card_status ADD CONSTRAINT fk_credit_card FOREIGN KEY ...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.219 sec

0=acceptada / 1=no accepta. Si la suma de les 3 ultimes transaccions és 3, llavors la targeta es marca com declinada.

Genera la següent consulta: Quantes targetes estan actives?

```

1 /* Sprint 4: consulta nivell 2 - exercici 1*/
2 /* Genera la següent consulta: Quantes targetes estan actives? */
3
4 select count(*) as targetes_actives from credit_card_status where status=0

```

#	Time	Action	Message	Duration / Fetch
1	12:33:21	select count(*) as targetes_actives from credit_card_status where status=0	1 row(s) returned	0.000 sec / 0.000 sec

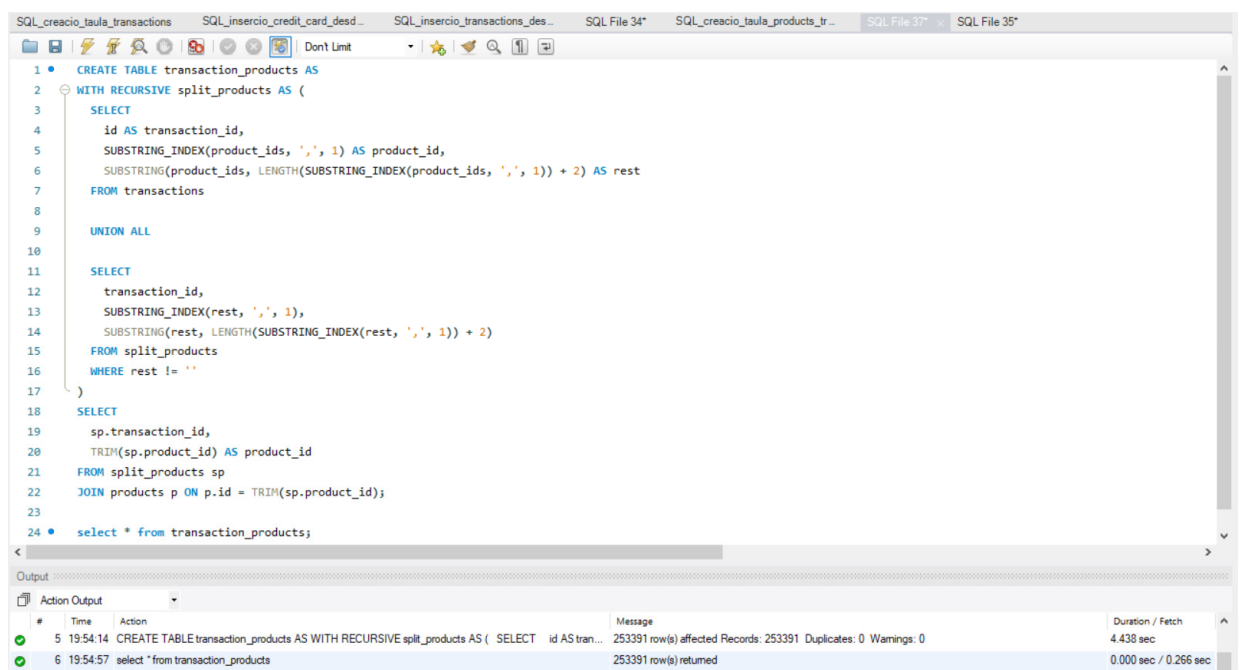
NIVELL 3: Exercici 1

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de *transaction* tens *product_ids*.

Al haver-hi un camp dins la taula *transactions* anomenat *product_ids* el qual conte varis productes per transacció, hem de crear una taula nova perquè relacioni *n* productes per *una* transacció. Anomeno a la taula *transaction_products* i la creo usant funcions de tractament de text i de recursivitat per treure tots els *id* de producte i separar-los en aquesta taula nova amb el seu respectiu *id* de transacció.

Creació nova taula *transaction_products*. 2 opcions:

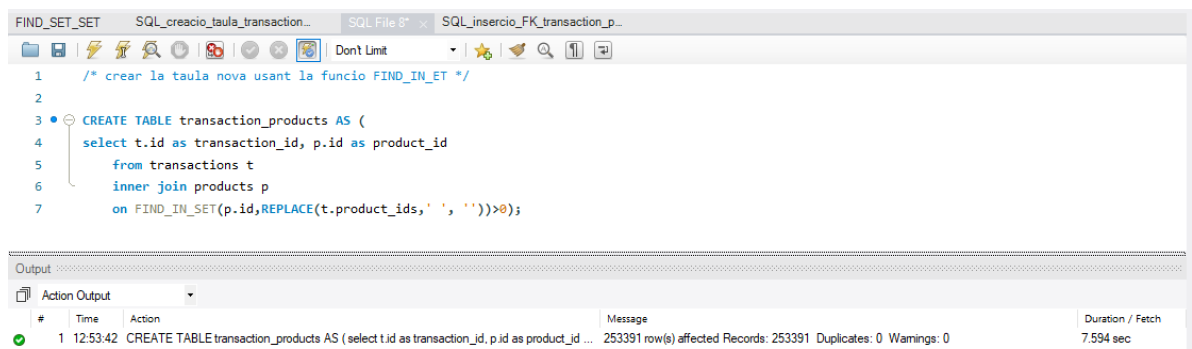
OPCIÓ 1: Sense usar la funció **FIND_IN_SET**



```
1 CREATE TABLE transaction_products AS
2 WITH RECURSIVE split_products AS (
3     SELECT
4         id AS transaction_id,
5         SUBSTRING_INDEX(product_ids, ',', 1) AS product_id,
6         SUBSTRING(product_ids, LENGTH(SUBSTRING_INDEX(product_ids, ',', 1)) + 2) AS rest
7     FROM transactions
8
9     UNION ALL
10
11     SELECT
12         transaction_id,
13         SUBSTRING_INDEX(rest, ',', 1),
14         SUBSTRING(rest, LENGTH(SUBSTRING_INDEX(rest, ',', 1)) + 2)
15     FROM split_products
16     WHERE rest != ''
17 )
18 SELECT
19     sp.transaction_id,
20     TRIM(sp.product_id) AS product_id
21 FROM split_products sp
22 JOIN products p ON p.id = TRIM(sp.product_id);
23
24 select * from transaction_products;
```

#	Time	Action	Message	Duration / Fetch
5	19:54:14	CREATE TABLE transaction_products AS WITH RECURSIVE split_products AS (SELECT id AS tran...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	4.438 sec
6	19:54:57	select * from transaction_products	253391 row(s) returned	0.000 sec / 0.266 sec

OPCIÓ 2: Usant la funció **FIND_IN_SET**



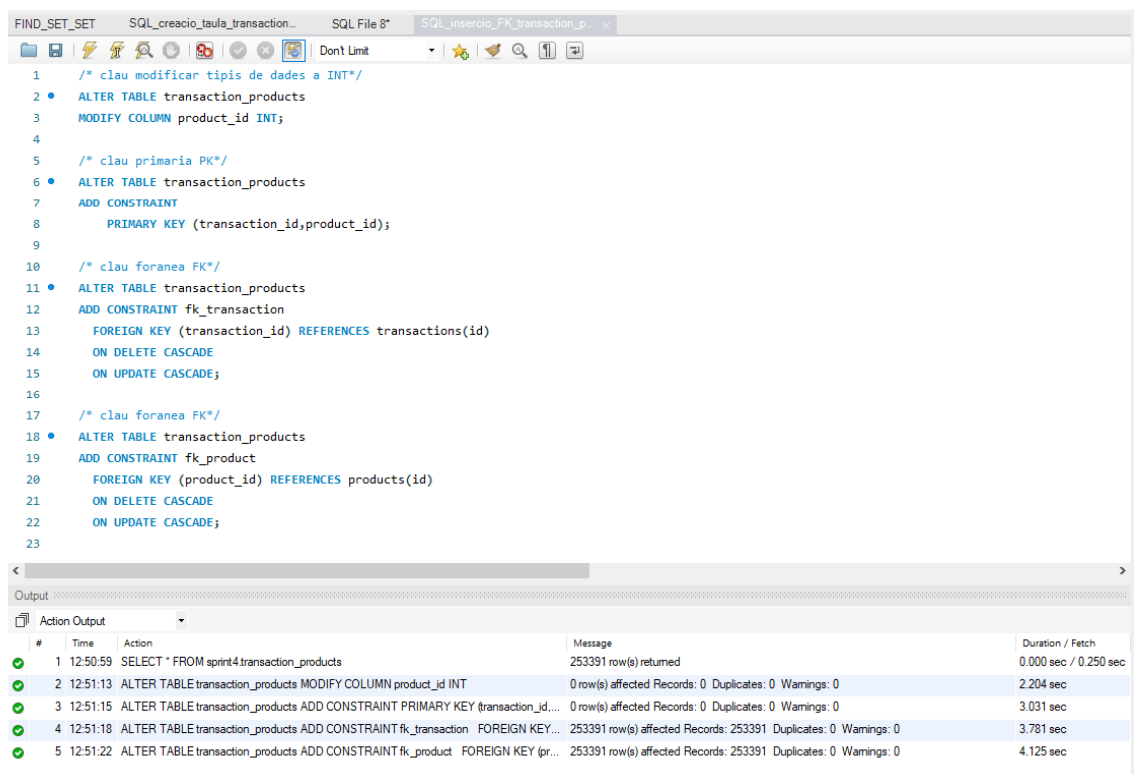
```
1 /* crear la taula nova usant la funcio FIND_IN_ET */
2
3 CREATE TABLE transaction_products AS (
4   select t.id as transaction_id, p.id as product_id
5   from transactions t
6   inner join products p
7   on FIND_IN_SET(p.id, REPLACE(t.product_ids, ', ', ''))>0);
```

Output

#	Time	Action	Message	Duration / Fetch
1	12:53:42	CREATE TABLE transaction_products AS (select t.id as transaction_id, p.id as product_id ...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	7.594 sec

CREACIÓ de les relacions (CONSTRAINT):

Creo la clau primària(PK) i les claus foranes(FK) sobre els dos camps. Modifico a INT el camp *product_id* per que coincideixi amb el INT del camp *id* de la taula *products* i fer la relació entre ells.



```
1 /* clau modificar tipis de dades a INT*/
2 ALTER TABLE transaction_products
3 MODIFY COLUMN product_id INT;
4
5 /* clau primaria PK*/
6 ALTER TABLE transaction_products
7 ADD CONSTRAINT
8 PRIMARY KEY (transaction_id, product_id);
9
10 /* clau foranea FK*/
11 ALTER TABLE transaction_products
12 ADD CONSTRAINT fk_transaction
13 FOREIGN KEY (transaction_id) REFERENCES transactions(id)
14 ON DELETE CASCADE
15 ON UPDATE CASCADE;
16
17 /* clau foranea FK*/
18 ALTER TABLE transaction_products
19 ADD CONSTRAINT fk_product
20 FOREIGN KEY (product_id) REFERENCES products(id)
21 ON DELETE CASCADE
22 ON UPDATE CASCADE;
23
```

Output

#	Time	Action	Message	Duration / Fetch
1	12:50:59	SELECT * FROM sprint4.transaction_products	253391 row(s) returned	0.000 sec / 0.250 sec
2	12:51:13	ALTER TABLE transaction_products MODIFY COLUMN product_id INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	2.204 sec
3	12:51:15	ALTER TABLE transaction_products ADD CONSTRAINT PRIMARY KEY (transaction_id, product_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	3.031 sec
4	12:51:18	ALTER TABLE transaction_products ADD CONSTRAINT fk_transaction FOREIGN KEY (transaction_id) REFERENCES transactions(id) ON DELETE CASCADE ON UPDATE CASCADE	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	3.781 sec
5	12:51:22	ALTER TABLE transaction_products ADD CONSTRAINT fk_product FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE ON UPDATE CASCADE	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	4.125 sec

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

The screenshot shows a database query tool interface. The top pane contains a SQL query:

```
1 /* Sprint 4: consulta nivell 3 - exercici 1*/
2 /* crear subconsulta que mostri nombre de vegades que s'ha venut cada producte */
3
4 select product_name as Producte, count(product_id) as Total_Ventes from products p
5 inner join transaction_products tp on tp.product_id = p.id
6 group by Producte
```

The bottom pane displays the 'Result Grid' with the following data:

Producte	Total_Ventes
Winterfell	10071
the duel	10059
dooku solo	10022
Karstark Dorne	7694
skywalker ewok	7607
Karstark warden	7563
Niraulf Stannis	7470

The 'Output' pane shows the execution details: 13:44:05, select product_name as Producte, count(product_id) as Total_Ventes from products p inn..., 70 row(s) returned, Duration / Fetch: 0.375 sec / 0.000 sec.

Diagrama de la BBSS en forma d'estrella o floc de neu:

